

Calculus: Derivatives, Differentiation Rules & Tangent Lines

Grade 11–12 | Worked examples and practice reference

Topics Covered in This Session

Graphical representation of a derivative · Tangents and normals · Maxima and minima (previewed)

1 The Derivative as a Slope

Slope of a Line

For two points (x_1, y_1) and (x_2, y_2) on a line, the slope m equals the tangent of the angle θ the line makes with the x -axis:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta.$$

Definition of the Derivative

For a curve $y = f(x)$, the derivative is defined as the limit of the average rate of change as the interval h shrinks to zero:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Note: This is exactly the same idea as slope – the derivative at a point gives the slope of the curve (i.e. of its tangent line) at that point: $f'(x) = \frac{y_2 - y_1}{x_2 - x_1} = m = \text{slope}$.

2 Differentiation Rules

- | | | |
|-----|--|--------------------------|
| i | $\frac{d}{dx}(x^n) = nx^{n-1}$ | Power rule |
| ii | $\frac{d}{dx}(c) = 0$, c is a constant | Derivative of a constant |
| iii | $\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx}$ | Product rule |
| iv | $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ | Quotient rule |

Chain Rule

For a composite function $h(x) = (u \circ v)(x)$:

$$h'(x) = \frac{d}{dx} [(u \circ v)(x)] = \frac{d}{dx} [u(v(x))] \cdot \frac{d}{dx} [v(x)].$$

$$\text{vi} \quad y = e^x \Rightarrow y' = e^x \quad (\text{Exponential rule})$$

Trigonometric Derivatives

$$\begin{array}{ll} \text{vii} \quad \frac{d}{dx}(\sin x) = \cos x & \text{x} \quad \frac{d}{dx}(\sec x) = \sec x \tan x \\ \text{viii} \quad \frac{d}{dx}(\cos x) = -\sin x & \text{xi} \quad \frac{d}{dx}(\cot x) = -\csc^2 x \\ \text{ix} \quad \frac{d}{dx}(\tan x) = \sec^2 x & \text{xii} \quad \frac{d}{dx}(\csc x) = -\csc x \cot x \end{array}$$

3 Separation of Constants

Rule

When differentiating $f(ax)$, the constant a is carried straight through:

$$\frac{d}{dx} f(ax) = a f'(ax).$$

Example

$$\frac{d}{dx} \sin(7x) = 7 \cos(7x).$$

4 Worked Examples

(a) Power rule

$$y = x^5 \Rightarrow \frac{dy}{dx} = 5x^4$$

(b) A pure constant differentiates to zero

$$y = c^5 \text{ (} c \text{ constant)} \Rightarrow \frac{dy}{dx} = 0$$

(c) Product rule

$$y = x^3(1 + x)$$

$$y' = (1 + x)(3x^2) + 1 \cdot x^3 = 3x^3 + 3x^2 + x^3 = 4x^3 + 3x^2$$

(d) Quotient rule

$$y = \frac{1 + x}{x^2}$$

$$y' = \frac{x^2(1) - (1 + x)(2x)}{x^4} = \frac{x^2 - 2x - 2x^2}{x^4} = \frac{-x^2 - 2x}{x^4} = -\frac{x + 2}{x^3}$$

(e) Chain rule

$$y = (1 + x)^5$$

$$y' = 5(1 + x)^4 \cdot \frac{d}{dx}(1 + x) = 5(1 + x)^4$$

5 Tangent Lines

Definition

A **tangent** is a line that touches a curve at exactly one point. For $y = x^2$, the derivative is $2x$, so the slope of the tangent at any point x is given by:

$$y = x^2 \Rightarrow \frac{dy}{dx} = 2x.$$

Point–Slope Form of a Tangent

The equation of the tangent to a curve at the point (x_1, y_1) is:

$$(y - y_1) = m(x - x_1)$$

where the slope m is the derivative evaluated at that point:

$$m = \left[\frac{dy}{dx} \right]_{(x_1, y_1)}.$$

Worked Example: Tangent to $y = e^x$ at $(1, 1)$

Find the equation of the tangent to $y = e^x$ at the point $(1, 1)$.

Find the slope m by evaluating the derivative at $(1, 1)$:

$$m = \left[\frac{dy}{dx} \right]_{(1,1)} = \left[\frac{d}{dx} e^x \right]_{(1,1)} = [e^x]_{(1,1)} = e^1 = e.$$

Substitute $m = e$ back into the point-slope equation and simplify:

$$(y - 1) = e(x - 1) \Rightarrow y = ex - e + 1 \Rightarrow y = ex + (1 - e).$$

Result: the tangent to $y = e^x$ at $(1, 1)$ is $y = ex + (1 - e)$.

